

# Hierarchical Clustering on United States Weather Data

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## 1. Introduction

Hierarchical clustering is an effective method of machine learning that reveals how data points can be grouped at multiple levels utilizing a plethora of numeric features. Among many different fields and research areas, weather data is a sector that contains hundreds of different types of data from thousands of places around the world. This project seeks to utilize the different types of data provided by hierarchical clustering and weather data presented by the National Oceanic and Atmospheric Administration (NOAA) [1] to cluster each of the 50 states into different climate/geographical regions. This project also compares the overall accuracy of hierarchical clustering methods to the kmeans clustering model. PCA was also used to determine the principal components of what clusters the station data.

## 2. Data Exploration

The dataset used for this project was collected from the NOAA related group, the National Centers for Environmental Information [1]. They provide data from hundreds of thousands of weather monitoring stations across multiple countries. In particular, we chose one weather station for every state in the United States of America from January 1, 2021 through December 31, 2021. Each station was selected based on multiple criteria; it should be located within a large city within that state, preferably an international airport or a municipal airport if there were no international airports. This data does contain possible biased generalizations for each state such as a larger city may experience different weather due to increased pollution from cars and industry. Having each station in these situations, however, will create a more standardized environment compared to selecting some from large cities and others from secluded areas.

Features include but are not limited to: Average Dew Point Temperature, Average Dry Bulb Temperature, Average Relative Humidity, Average Sea Level Pressure, Average Station Pressure, Average Wet Bulb Temperature, Average Wind Speed, Cooling Degree Days, Departure From Normal Average, Heating Degree Days, Maximum Dry Bulb Temperature, Minimum Dry Bulb Temperature, Peak Wind Direction, Peak Wind Speed, Precipitation, Snow Depth, Snowfall, Sustained Wind Direction, Sustained Wind Speed, Weather. (These features all have hourly, daily, and monthly values depending on the csv selected.)

Other nominal features : Station Number, Date, report type, source, sunrise, sunset.

Among these features, only a few of them are represented in monthly frequencies such as temperature average, single day temperature min, single day temperature max, single day

The project turned out to be a challenging one. It's good that you chose an original topic and were fairly independent in your analysis.

The writing is mostly ok.

Project/Data A  
Execution A-  
Writing A

greatest precipitation, total precipitation, and snowfall. Elements like humidity and wind speed were not included in a monthly variation but instead in the hourly and daily data sections. We chose to utilize the monthly data first and foremost in our analysis and modeling.

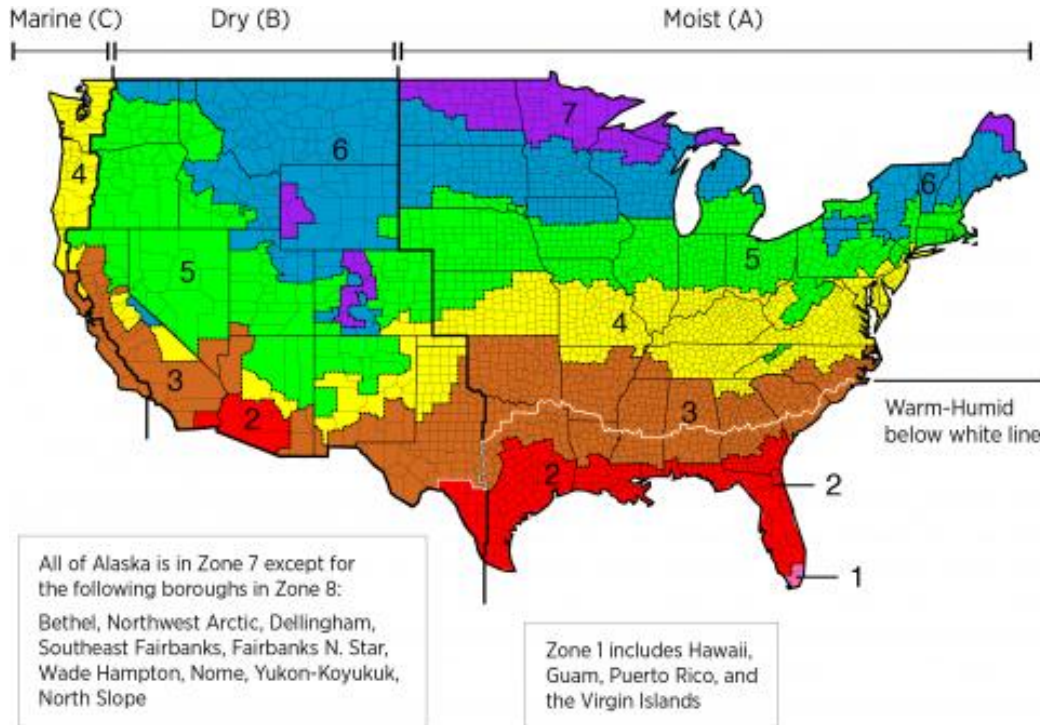
While exploring our data, not all stations were consistent. In the variable for station pressure, the state of Rhode Island and Arkansas were completely missing their station pressure readings as well as multiple other states missing a few months of station pressure data. This feature could be useful for analysis and some methods including using nearby states data can work for filling the gaps in data. However, with multiple stations missing this data, it is better to ignore the station pressure feature to increase integrity in the model.

### **3. Data Preparation**

The dataset had a feature which included a station number, but the state that the particular station resided in was not available in the dataset. In order to know what was getting clustered together, we added a state feature to the dataset by looking up the station number. This would make the results make much more sense to us and the reader.

The NOAA dataset was provided to us in csv format containing 10 stations worth of data. These csv's were combined into one large dataframe, the new state label was added, and then broken down into which feature(s) we were looking into. These additional csv's were broken down into each state being a row/observation and each column was a 1 month observation of that feature. This dataframe was crafted for the hierarchical clustering and kmeans implementation.

Using the map shown in figure 1 from the International Energy Conservation Code (IECC) [2], climate zone labels were added for each entree. The zones are labeled from 1-7, as well as A, B, or C for the Moist, Dry, and Marine zones on the map. Looking at multiple maps, weather station locations were pinpointed to their counties and assigned a label. In total, we ended up with 14 different total groupings. These different groupings were also added as a feature to the dataset, which is what we ended up using as our evaluation metric. We also decided to split these groupings into specifically the climate zone 1-7 groupings to see if a smaller number of clusters would prove to have a better classification rate. It is difficult to cluster 50 different items into 14 different clusters and have accurate results, so we hoped that having 7 clusters would improve the results.



**Figure 1.**

#### **4. Models and Results**

For determining model accuracy, we found each cluster using hierarchical clustering or kmeans and the group label was determined by the majority class between the observations in the cluster. All points that were misclassified then contributed to the overall misclassification rate.

Hierarchical clustering was utilized using multiple different distance metrics for calculating clusters. In this project, we utilized average, single, complete, centroid, and median centroid distances as well as the ward 1 algorithm that utilizes minimizing variance of clusters. Figure 2 below is an example dendrogram from the hierarchical clustering utilizing the average distance metric as well as a filled United States map based on predicted labels on figure 3 [3]. The remaining dendrograms will be in the appendix section but will be discussed in this section.

4.a 14 Groups

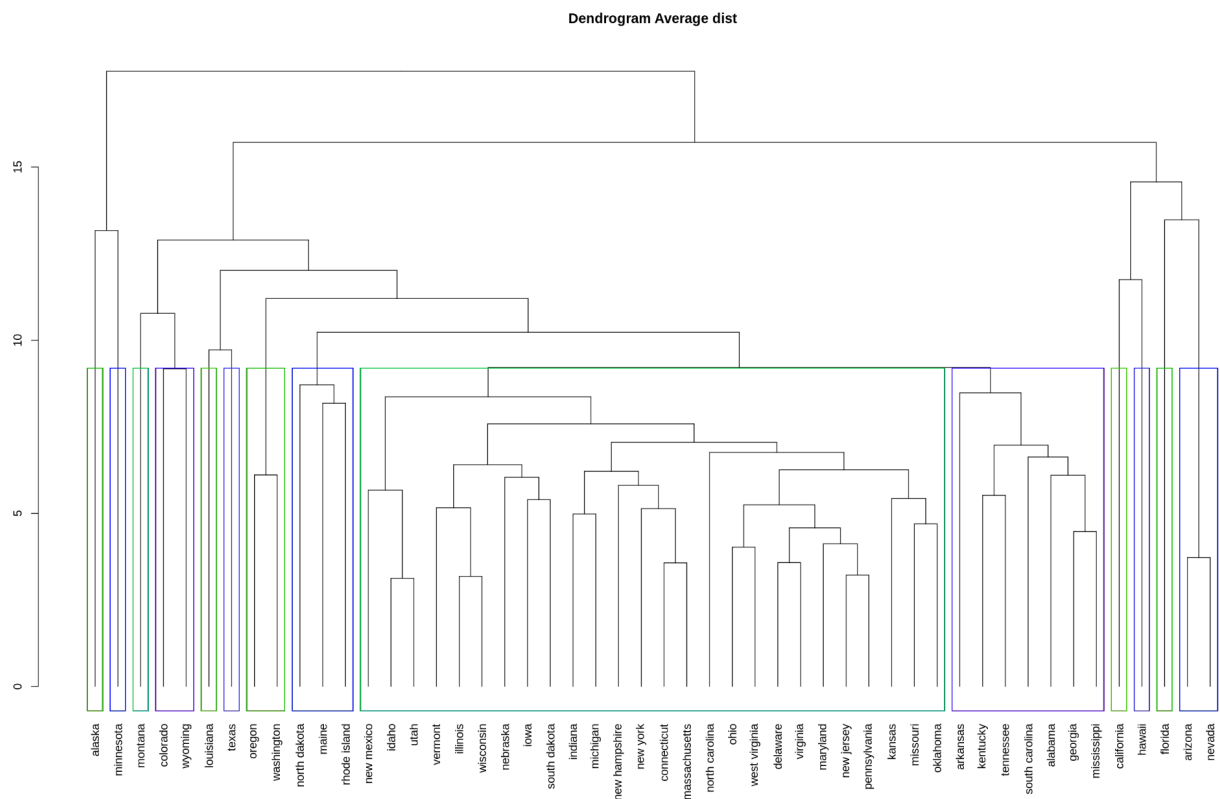
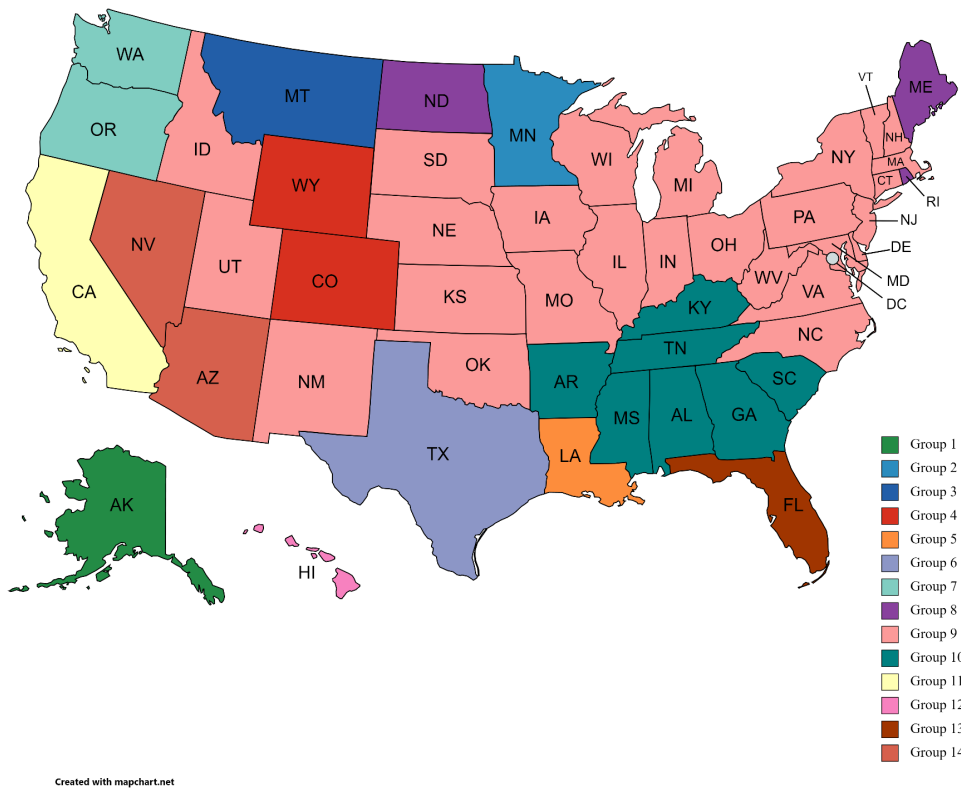


Figure 2.

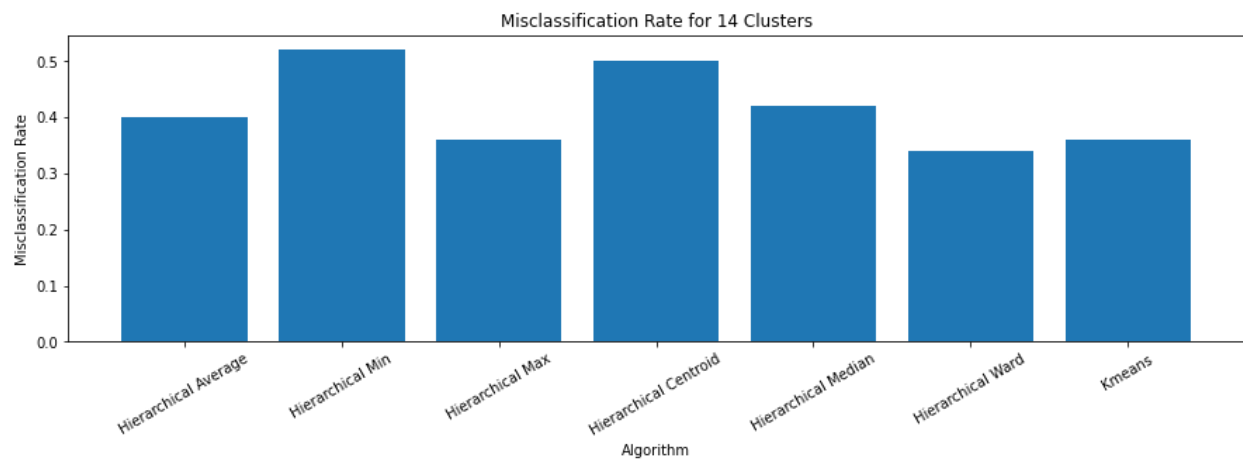


**Figure 3**

From the above dendrogram and corresponding map, the model clustered similar groups of states together. The cluster of states in the south are put together, the dry hot states of Nevada and Arizona are kept together and the colder states of Alaska and Minnesota are kept separate from other states. As well, a prevailing triumph of almost every model is it clustering the pacific northwest region of Washington and Oregon every time. The major issue with the above cluster and future clusters is its prevalence to clustering many of the moist region states into the same group even if they are in different climate zones. The average distance clustering is poorer than others as it puts New Mexico, Utah, and Idaho with the moist states even though they lack frequent precipitation. Table 1 shows the overall state classification as well as the misclassification rate for each model method.

**Table 1.**

| Method                  | Classified Correctly | Classified Incorrectly | Misclass Rate |
|-------------------------|----------------------|------------------------|---------------|
| Hierarchical - Average  | 30                   | 20                     | 0.4           |
| Hierarchical - Min      | 24                   | 26                     | 0.52          |
| Hierarchical - Max      | 32                   | 18                     | 0.36          |
| Hierarchical - Centroid | 25                   | 25                     | 0.5           |
| Hierarchical - Median   | 29                   | 21                     | 0.42          |
| Hierarchical - Ward     | 33                   | 17                     | 0.34          |
| Kmeans                  | 32                   | 18                     | 0.36          |

**Figure 4**

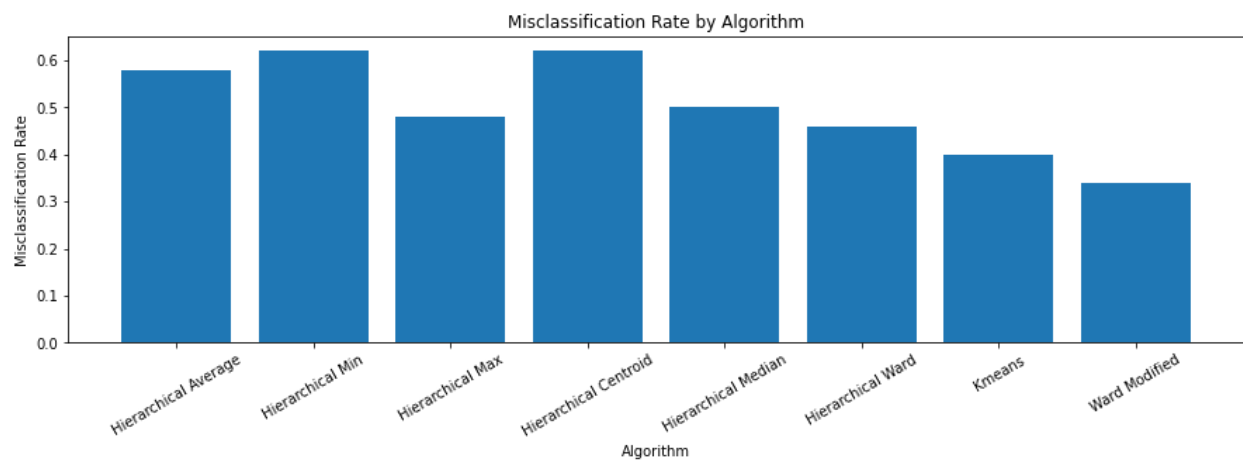
From the table, we can see that ward's algorithm in hierarchical clustering has the best misclassification rate at 34% closely followed by the max distance and kmeans clustering at 36%. The worst method was min distance with hierarchical clustering at 52% misclassification rate. This could be attributed to the large number of groups of 14, but could lowering it to the 7 climate groups without precipitation zones improve metrics?

## 4.b 7 Groups

Please see the appendix for the dendrogram results with 7 groups. Table 2 below shows the states classification

**Table 2**

| Method                                              | Classified Correctly | Classified Incorrectly | Misclass Rate |
|-----------------------------------------------------|----------------------|------------------------|---------------|
| Hierarchical - Average                              | 21                   | 29                     | 0.58          |
| Hierarchical - Min                                  | 19                   | 31                     | 0.62          |
| Hierarchical - Max                                  | 26                   | 24                     | 0.48          |
| Hierarchical - Centroid                             | 19                   | 31                     | 0.62          |
| Hierarchical - Median                               | 25                   | 25                     | 0.5           |
| Hierarchical - Ward                                 | 27                   | 23                     | 0.46          |
| Kmeans                                              | 30                   | 20                     | 0.4           |
| Hierarchical - Ward using only temperature features | 33                   | 17                     | 0.34          |



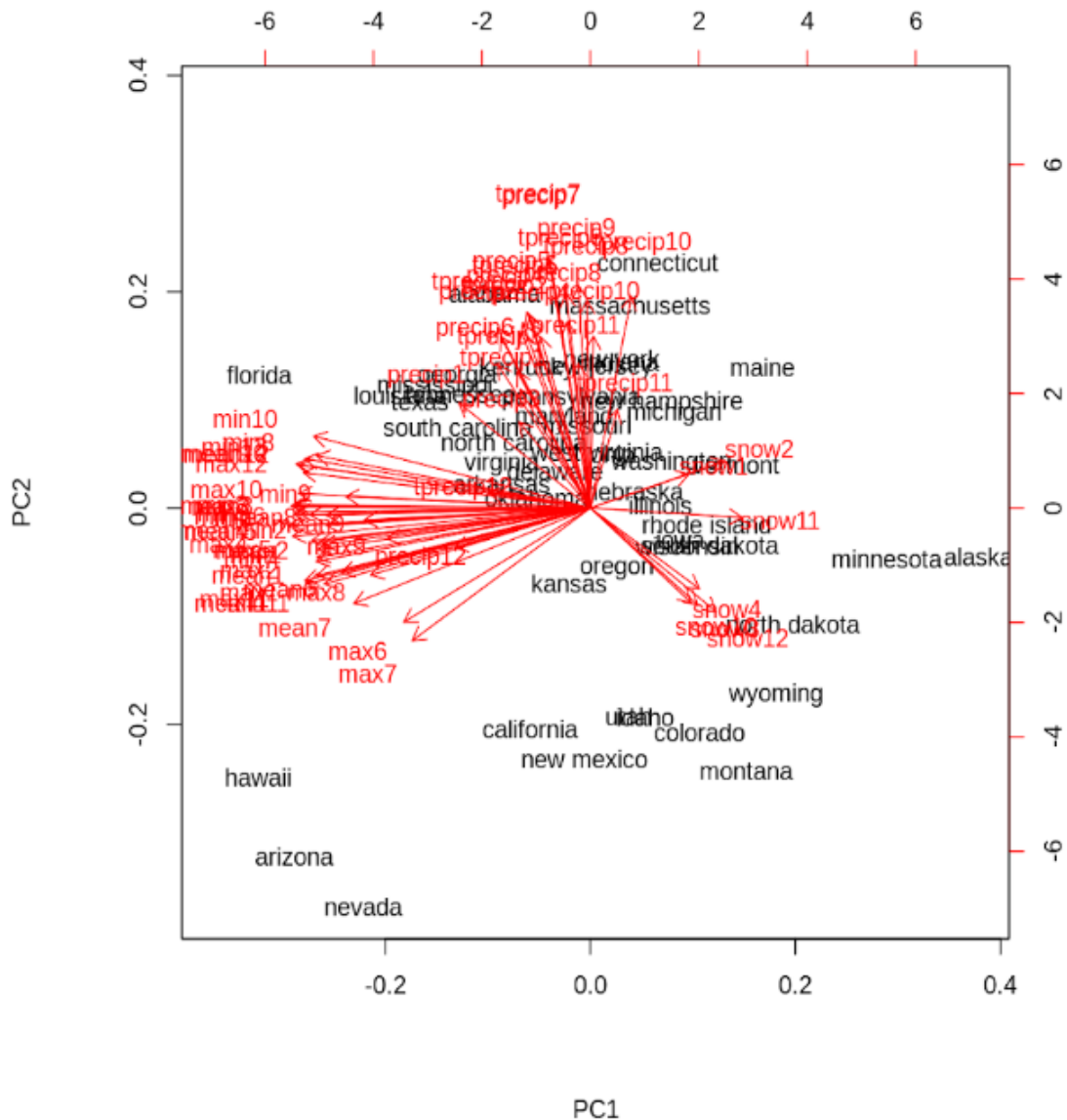
**Figure 5**

With 7 groups, the overall misclassification rate is higher than 14 groups. The best is Kmeans with a 40% misclassification rate followed by ward's algorithm at 46%. This could be a product of the features being used. The climate zones 1-7 reflect the overall temperature of each area which can be a product of the level of precipitation but that is not always the case. Therefore, giving the algorithm both temperature and precipitation features may be misleading for what it should be classifying. When applying Ward's algorithm to cluster features containing only temperature, the misclassification rate becomes 34%. Therefore, there is more of a linear relationship between temperature at the weather station and the climate zones 1-7. The precipitation may be good for clustering all 14 zones but including precipitation for 7 clusters misleads the algorithms more than it improves.

#### **4.c PCA**

The PCA algorithm was applied to the whole set of features including temperature, precipitation, and snowfall. Figure 6 below shows the biplot from the Principal Component Analysis.





**Figure 6**

For clarification, the features above are classified by the metric followed by the month it was taken (example, min8 is the minimum temperature value in the month of August for each state). From this, it shows that the first principal component is determined by temperature on one side and snowfall on the other. The negative side of principal component 1 correlates to high temperatures and positive side is lower temperatures. This is exemplified by Hawaii and Arizona towards the negative end on PC1 and Minnesota and Alaska towards the positive end. Principal component 2 is determined by precipitation with the dryer states of Arizona and Nevada on the negative ends and the wetter states of Connecticut and Florida on the positive side. Hawaii seems to be an outlier in terms of principal component 2 as it does receive precipitation but otherwise the PCA shows how these clustering algorithms are dividing the states.

## **5. Future Work**

Though this project touches on many elements of clustering and weather data, there are many elements we would like to explore after this. First involves expanding the dataset to multiple stations within each state. This could show the nuances that each state represents in the different climate zones as well as expand the dataset to improve classification rate. Secondly, we would like to include different features in our models. Unfortunately, only using monthly data greatly affected which data we could use (no humidity and no pressure). However, we could derive some features from the daily or hourly metrics as well as transformations of the monthly data. This could increase classification accuracy depending on the features we derive.

We explored separating the groups by less than 14 groups with the 7 temperature climate zones. This proved to reduce accuracy due to feature selection. In terms of future work, we will look to cluster the stations by the 7 temperature and 3 precipitation climate zones, however, there will need to be extra care in feature selection for this analysis.

## **6. Conclusion**

Overall, hierarchical clustering is adequate in clustering the 50 stations in terms of the climate zones. In particular, the methods of wards algorithm and kmeans overperformed in comparison to the other clustering methods. However, the misclassification rate was high for all methods of clustering. This is most likely a product of having many labels for only a few observations. Even though the misclassification rate tended to be high, the different groupings still represented the geographical regions of the United States well in the eye test.

This project taught us many different things inside and outside of the Statistical Machine Learning course. First and foremost, the concepts of PCA were solidified during this project. The task of not only performing PCA but utilizing it to understand clustering was lost on us during the class work but solidified after this project. Secondly, the methods of determining clusters and seeing which were most effective was solidified in this project. Lastly, we have worked with supervised learning for a majority of our Computer Science courses but this project focused more on interpreting unsupervised learning. This not only benefitted our learning but it also strengthened our Machine Learning professor's understanding of Hierarchical Clustering and R's machine learning methods.

## A. Bibliography

[1] National Centers for Environmental Information. “Data Tools: Local Climatological Data (LCD)” *National Oceanic and Atmospheric Administration*. [Online]. Available: <https://www.ncdc.noaa.gov/cdo-web/datatools/lcd>. [Accessed Oct 10, 2022].

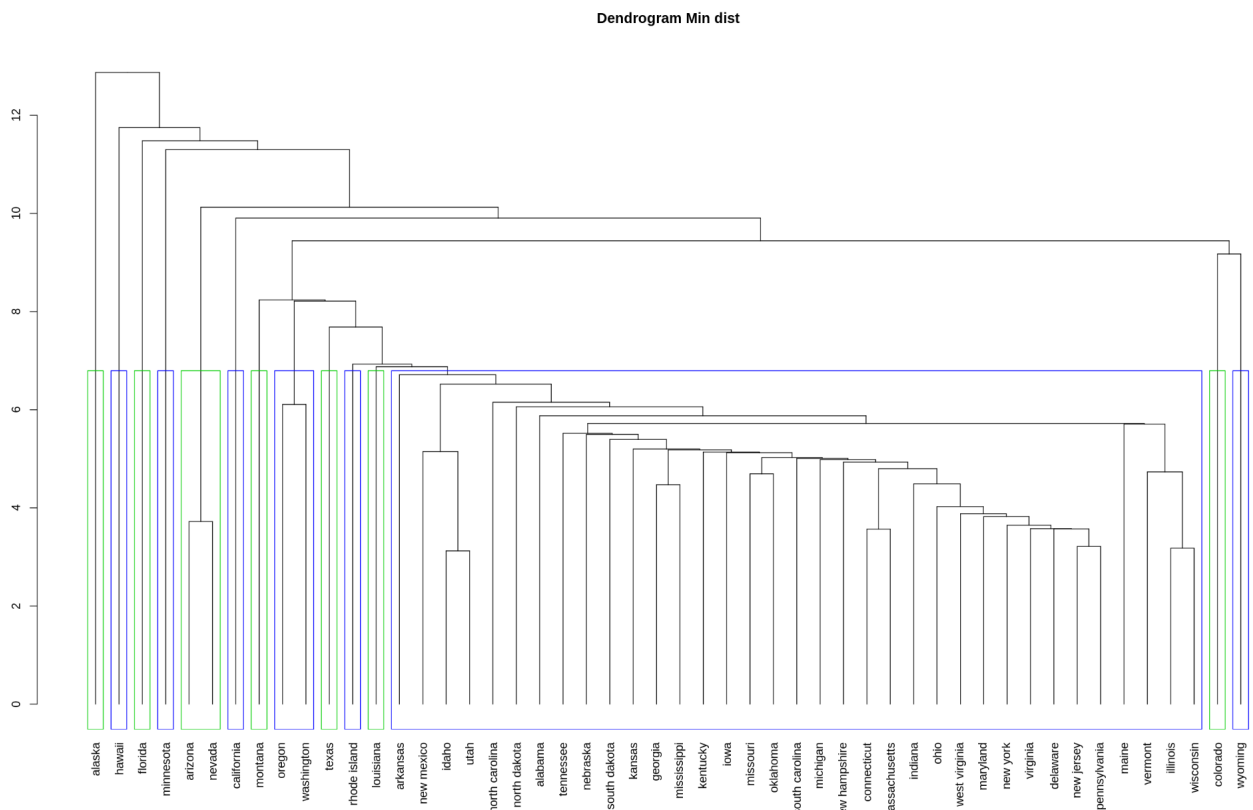
[2] Office of Energy Efficiency and Renewable Energy. “IECC climate zone map”. *International Energy Conservation Code*. [Online]. Available: <https://basc.pnnl.gov/images/iecc-climate-zone-map>. [Accessed Oct 25, 2022].

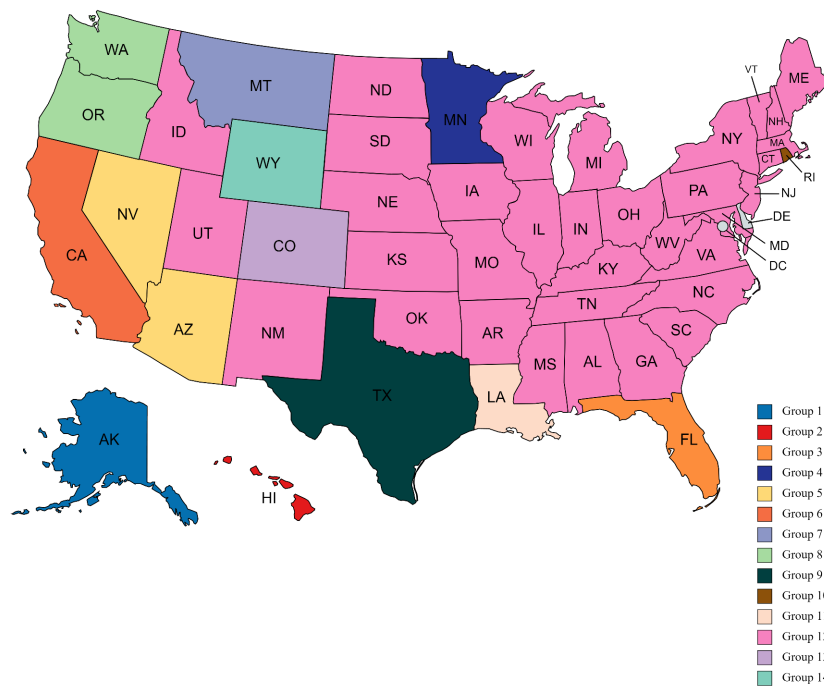
[3] MapChart. “United States”. *Map Chart* [Online] . Available: <https://www.mapchart.net/usa.html>. [Accessed Oct 25, 2022].

## B. Appendix

## 14 Groups:

### Minimum Distance

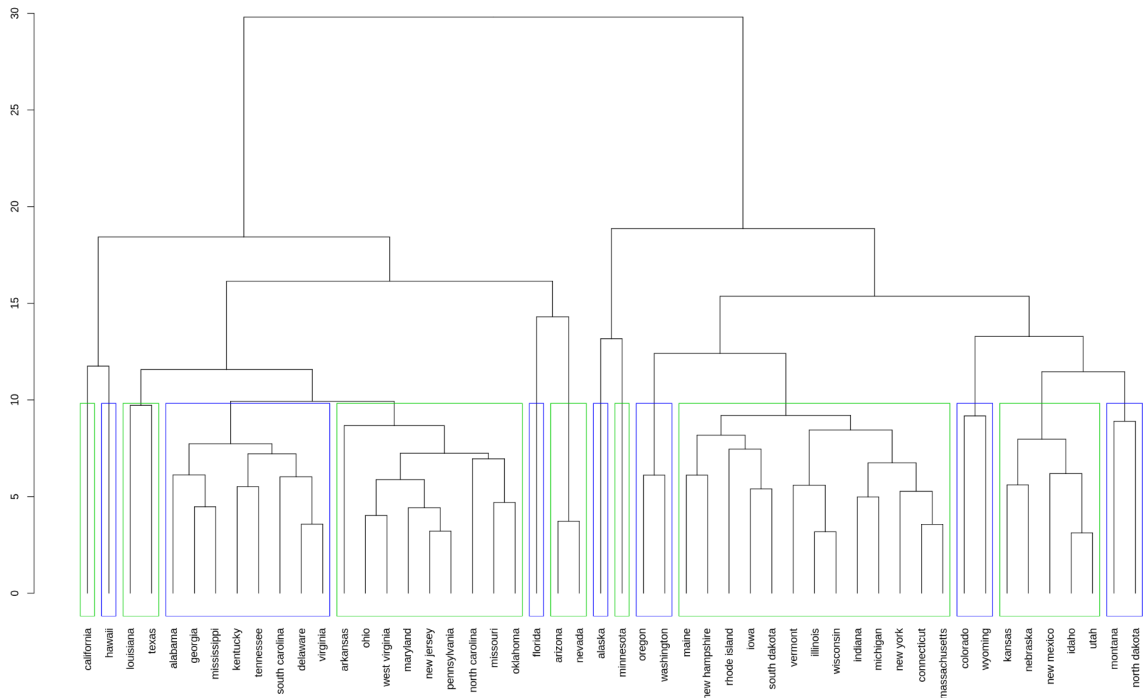


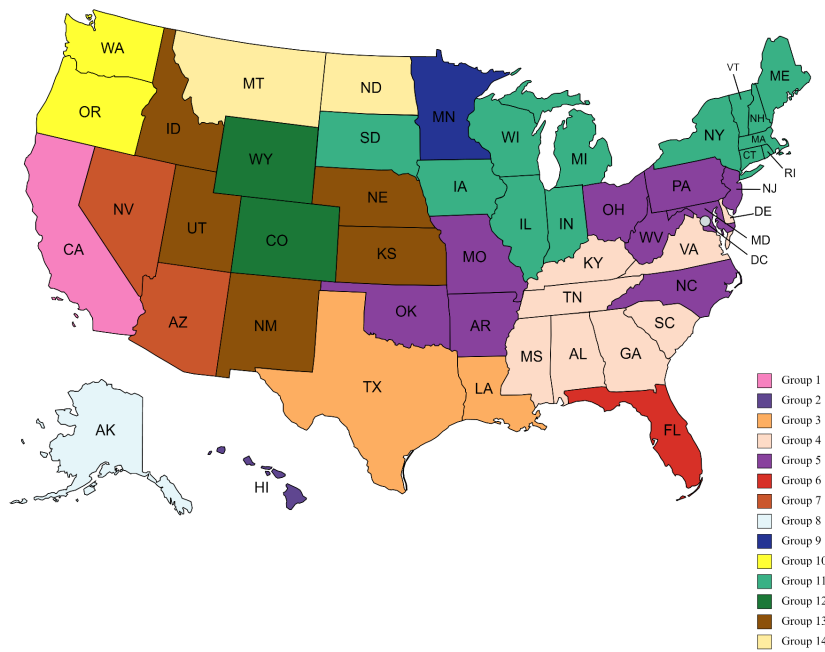


Created with mapchar.net

# Maximum Distance

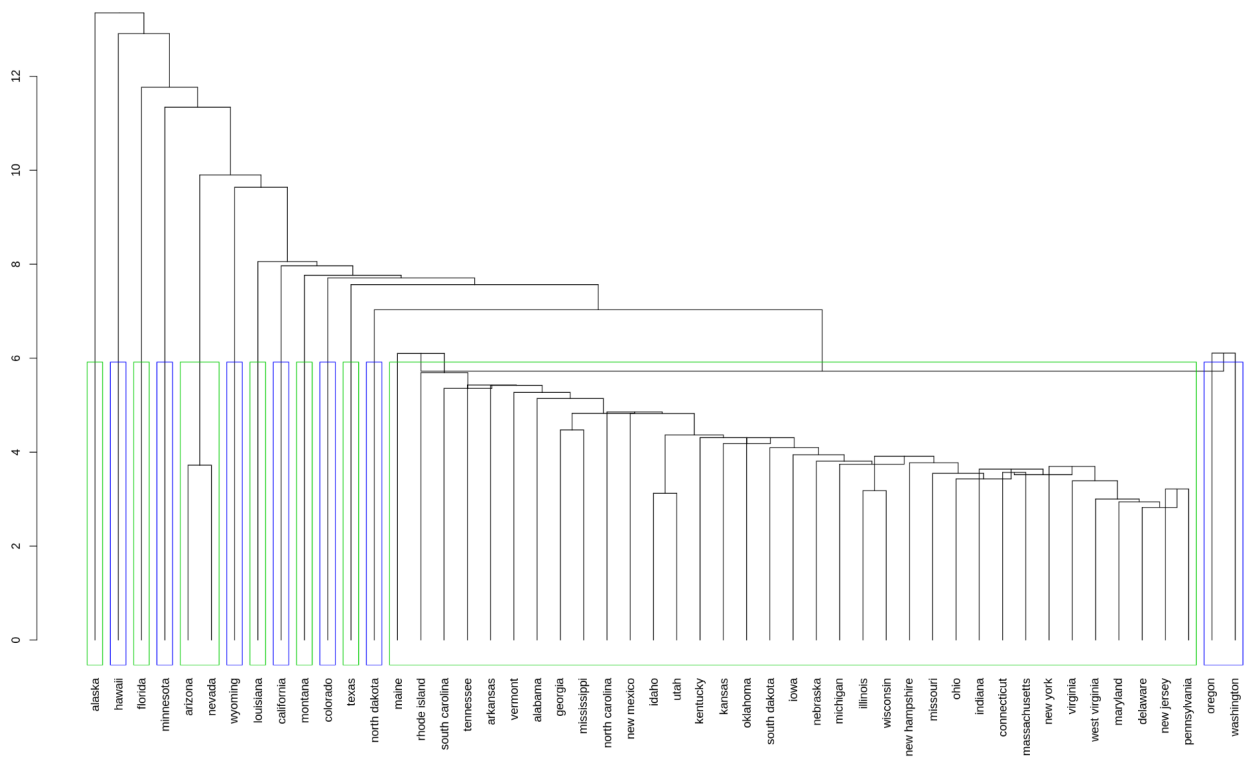
Dendrogram Max dist

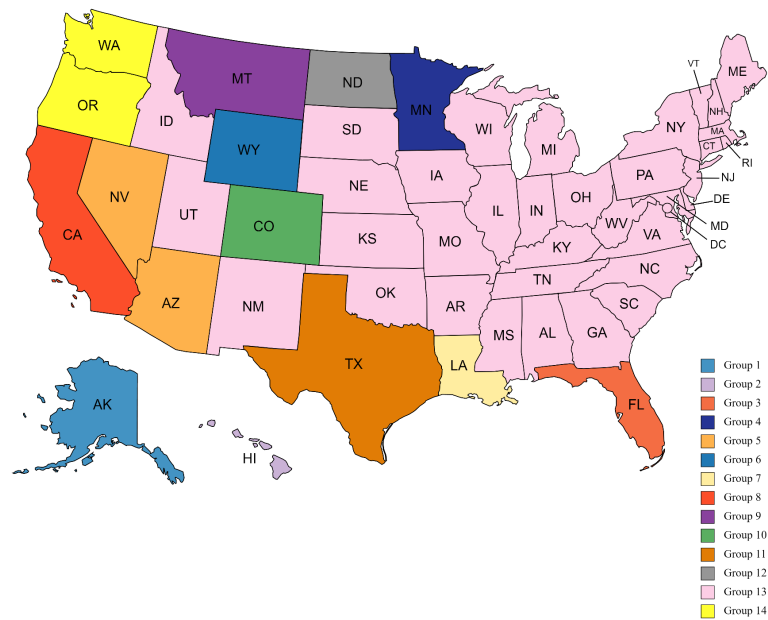




## Centroid Distance

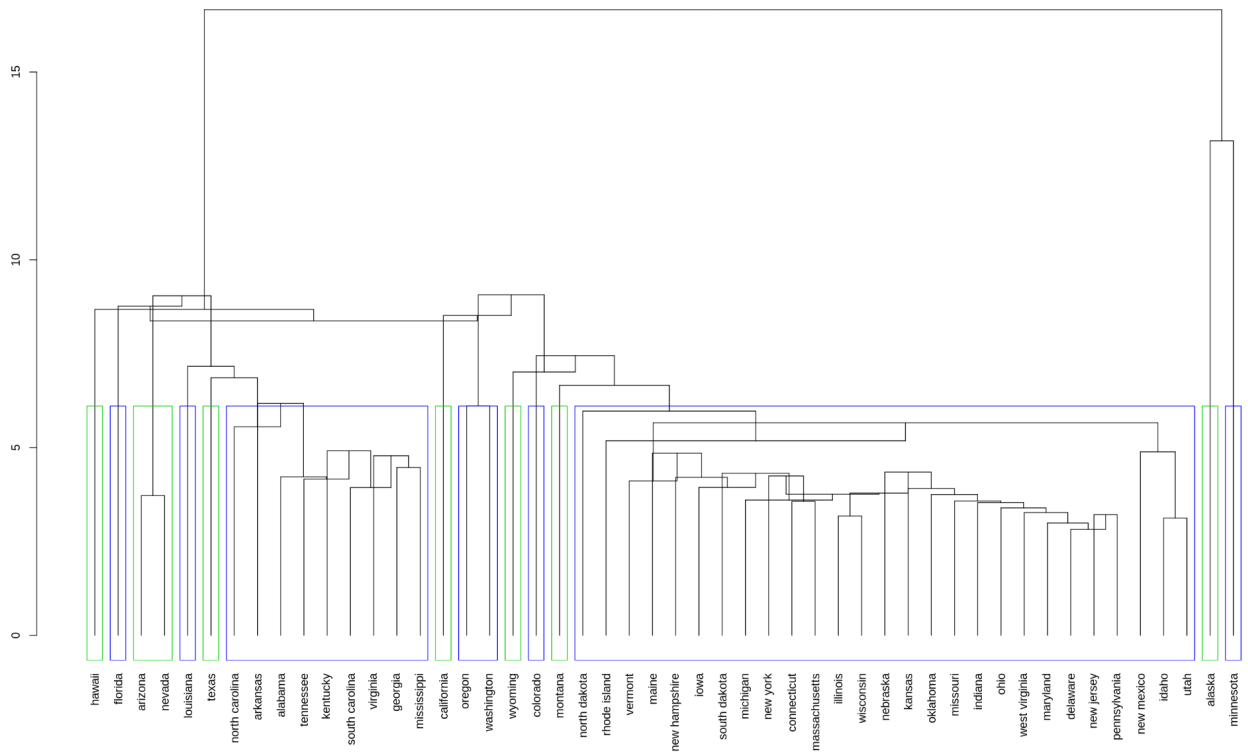
Dendrogram Centroid dist

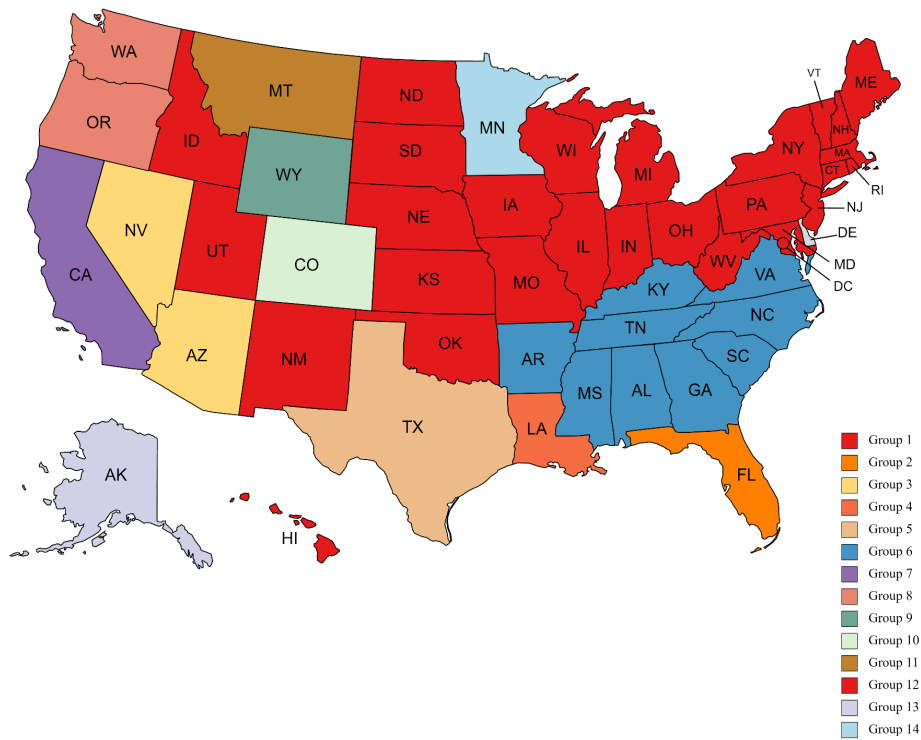




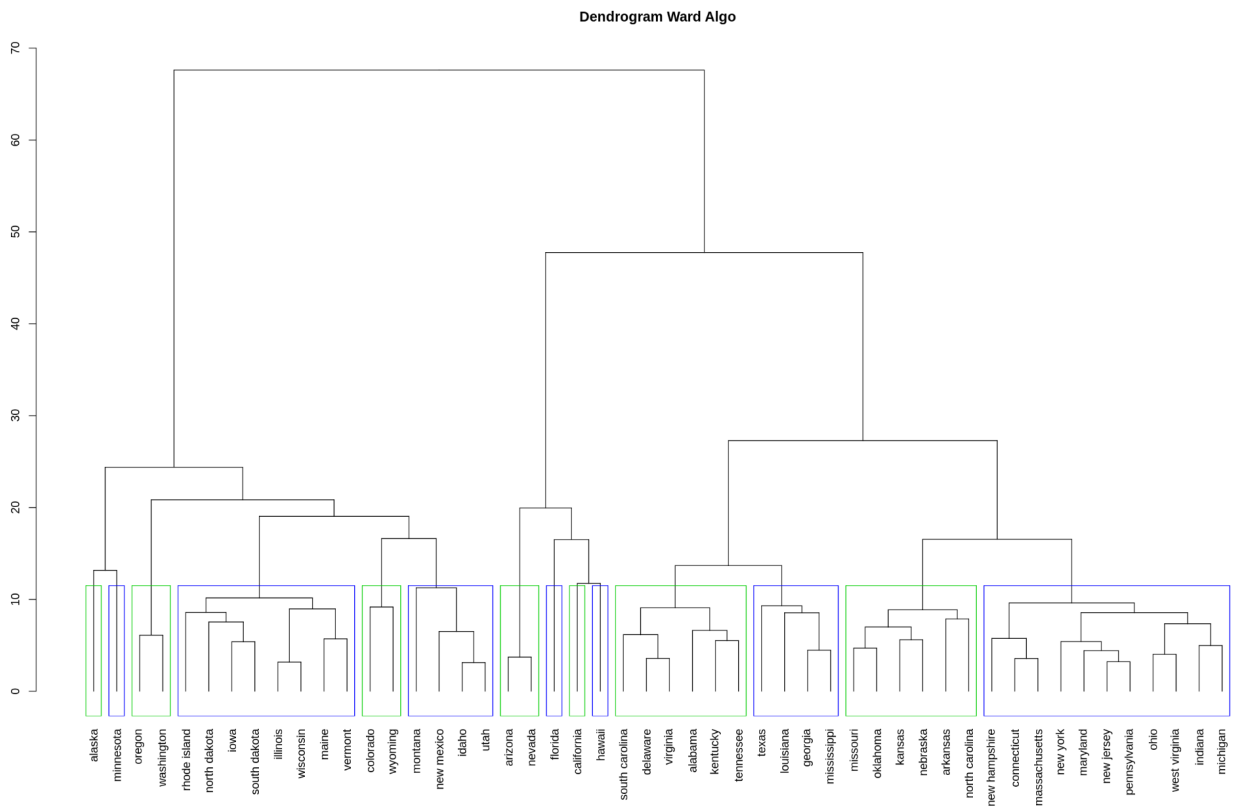
## Median Centroid

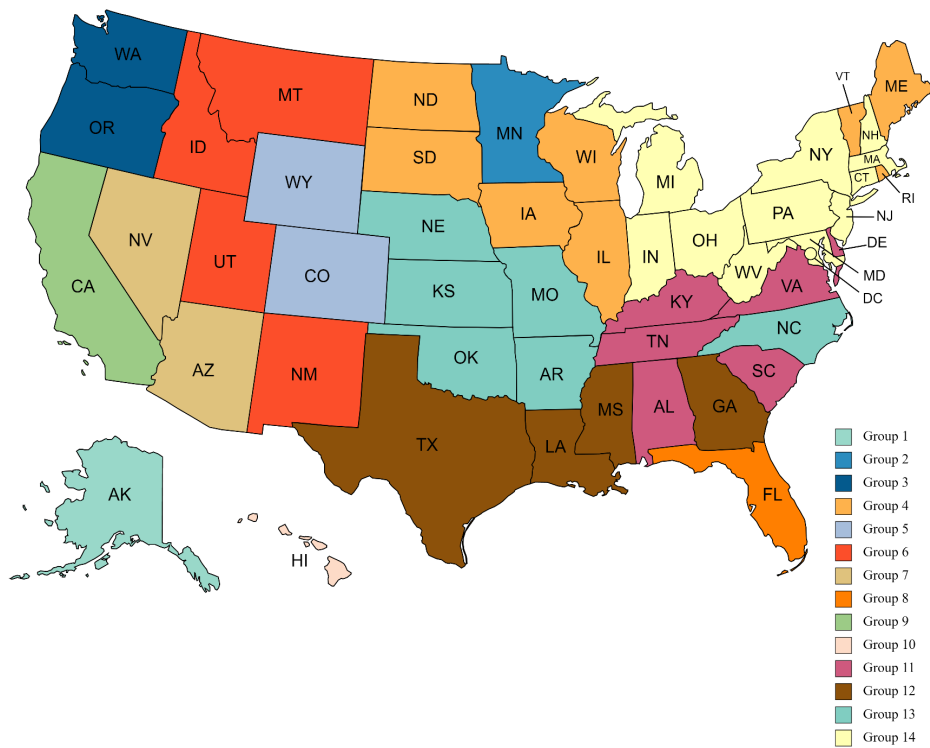
Dendrogram Median dist





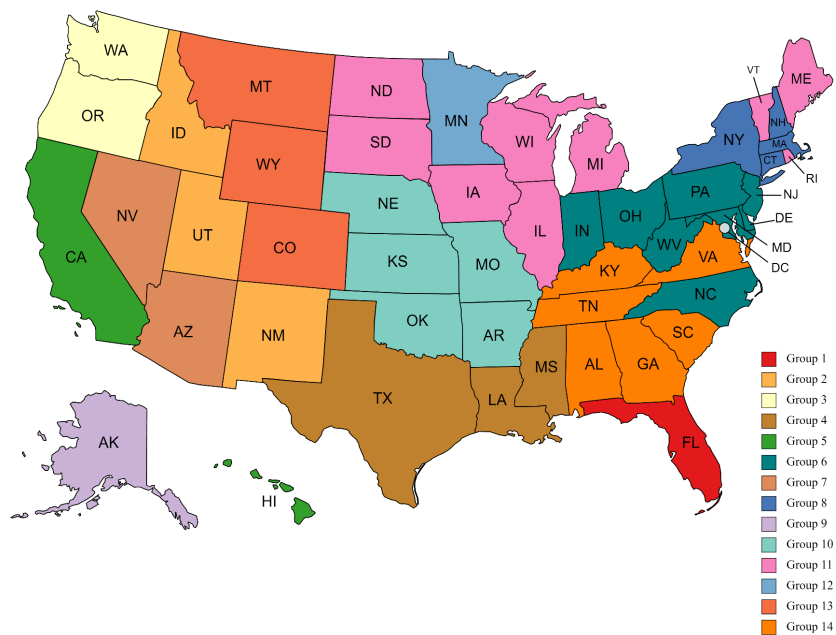
## Ward 1





Created with mapchart.net

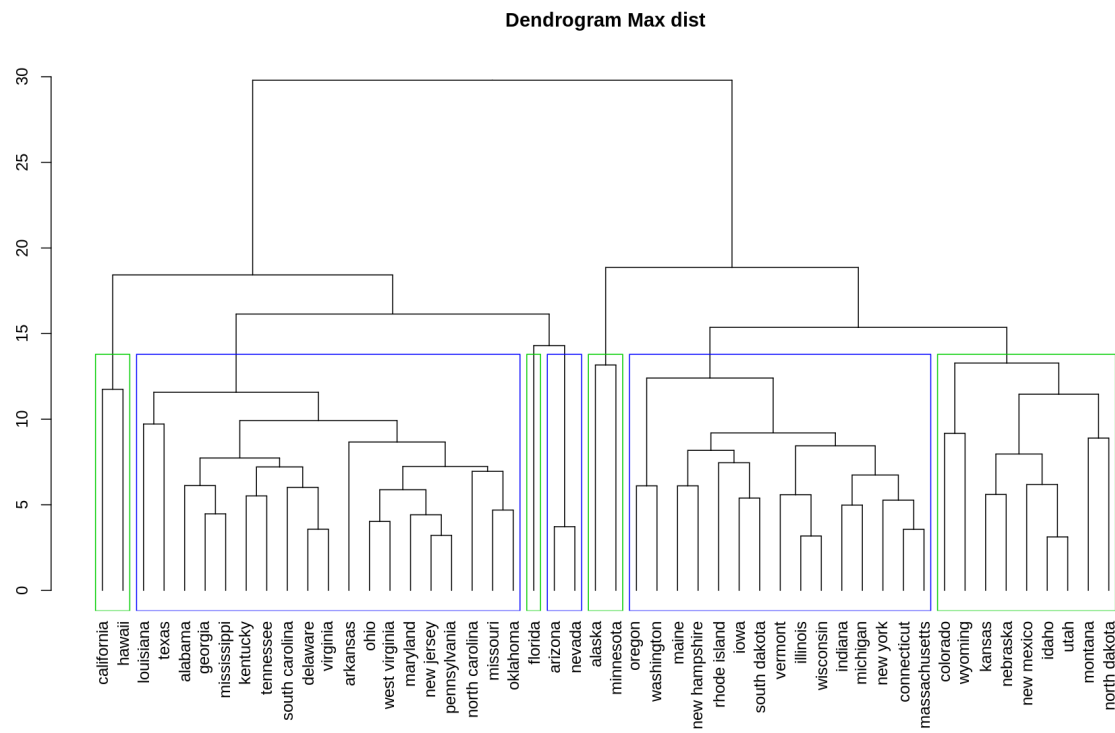
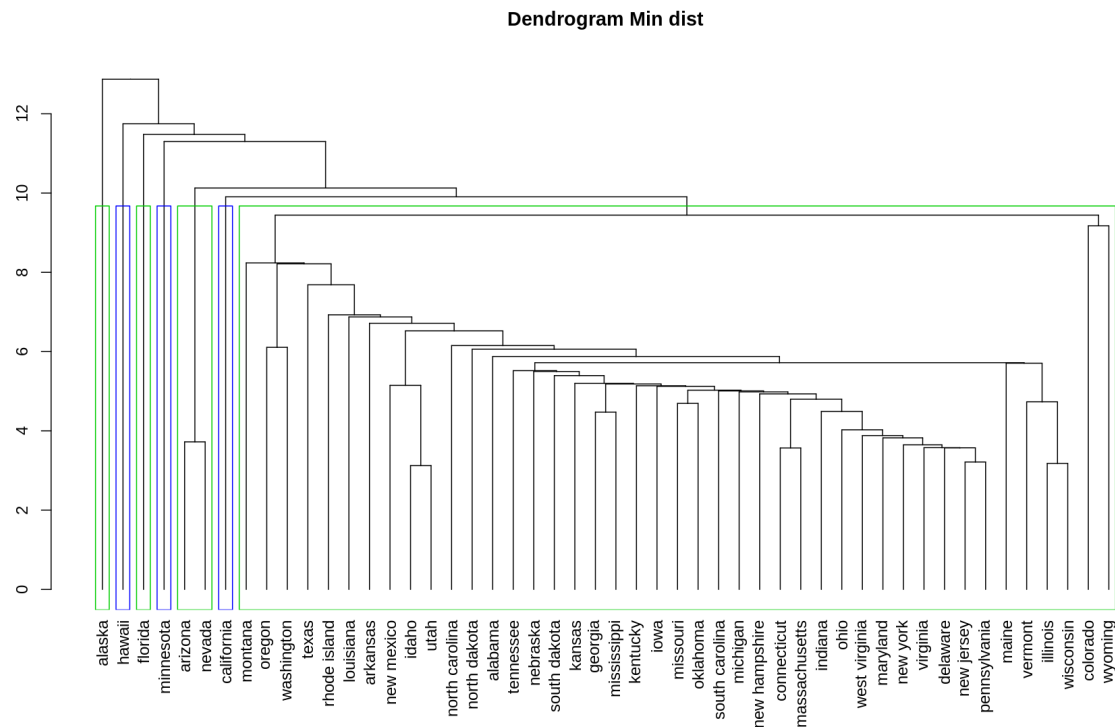
## Kmeans



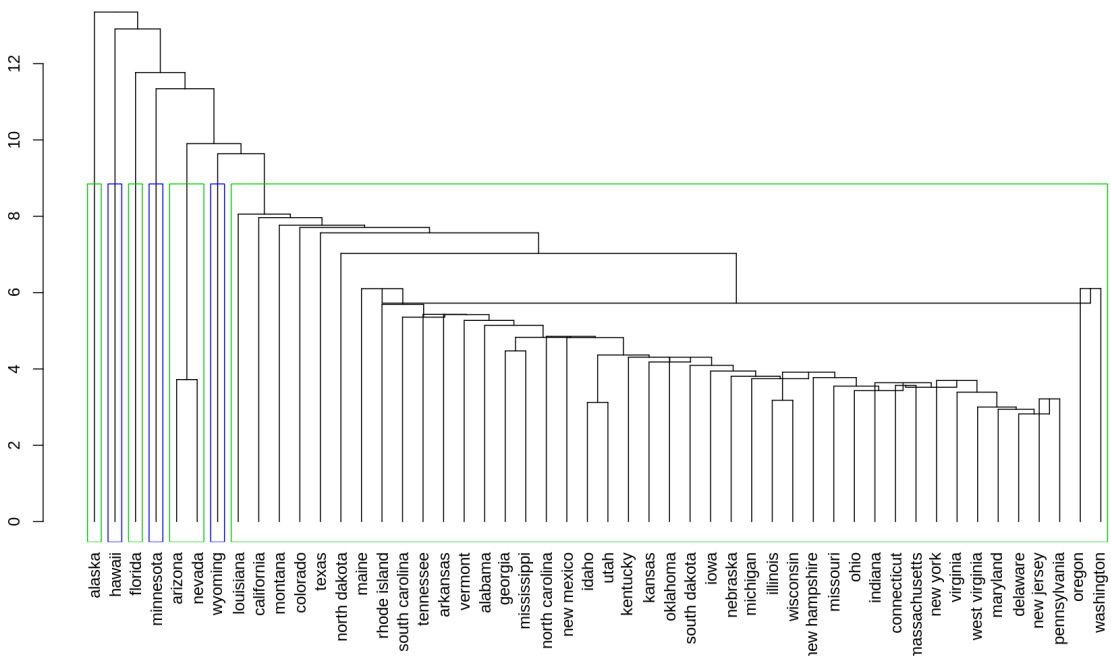
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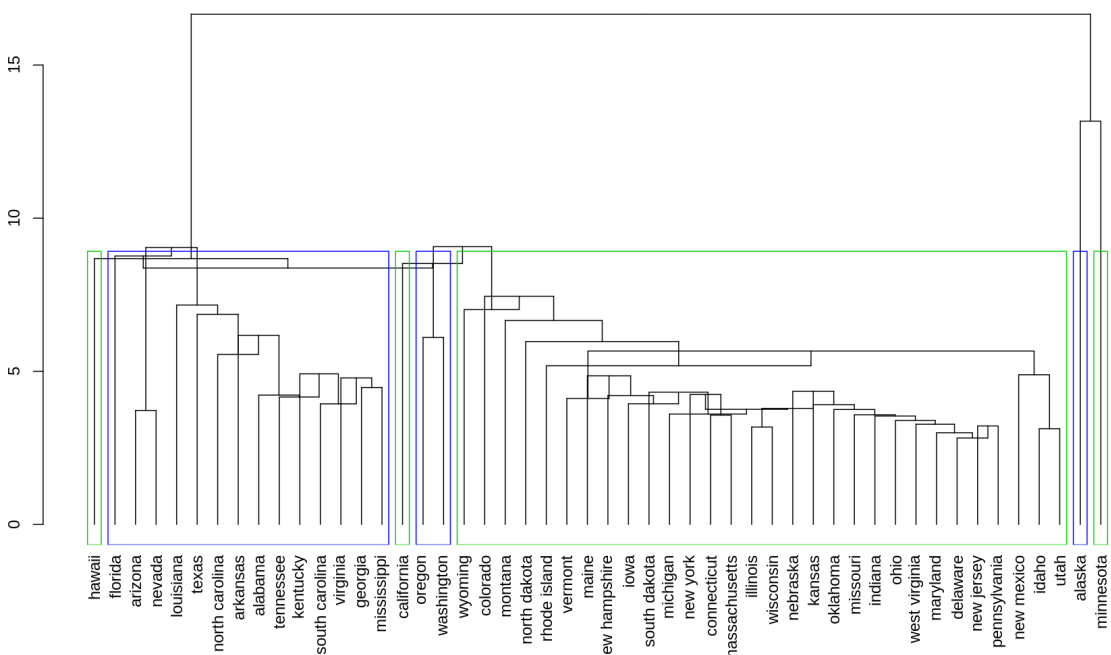
7 Groups:

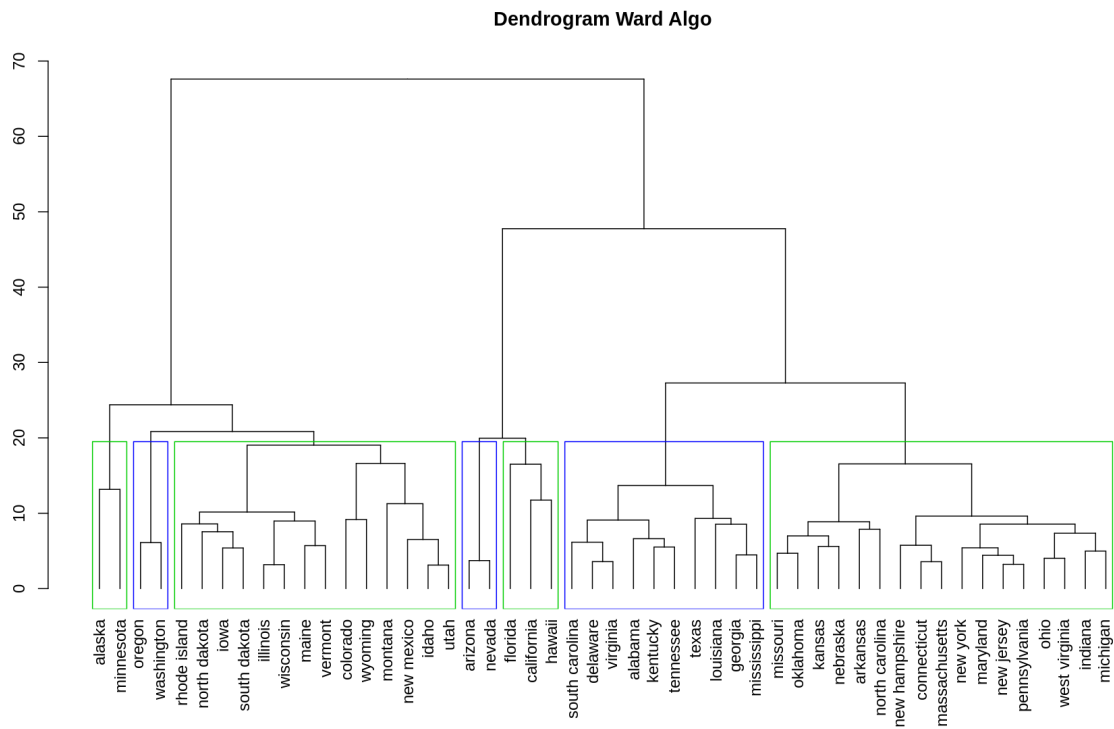


Dendrogram Centroid dist



Dendrogram Median dist





## Ward with Only Temperature

